

Firewall Change Request (CRF) — OpenShift 4.20 Air-gapped Cluster on vSphere (Revised)

Excludes 3 rows removed per review: (1) Master/worker ↔ Master/worker 9000-9999 host-level services, (2) Master/worker → DNS Server 53 name resolution, (3) Bastion → Master/worker 22 SSH troubleshooting. Source: gm_h_OCP-VM-Network. Destination category: App.

S.No	Source Zone / Device	Source IP/CIDR	Source Role	Dest Zone / Device	Destination IP	Destination Name	Ports	Protocol	Port-wise Description	Duration	Remarks
1	Master/worker	172.17.31.128/27	Master/worker	App	172.17.31.128/27	Master	6443, 22623, 22624	TCP	API server + MCS (all nodes call masters)	Temporary	
2	Master	172.17.31.128/27	Master	App	172.17.31.128/27	Master	2379-2380, 9978-9980	TCP	etcd client/peer + metrics (masters only)	Temporary	
3	Master/worker	172.17.31.128/27	Master/worker	App	172.17.31.128/27	Master/worker	10250-10259	TCP	Kubelet (all nodes) + controller-manager/scheduler (masters)	Temporary	
4	Master/worker	172.17.31.128/27	Master/worker	App	172.17.31.128/27	Master/worker	6081	UDP	OVN-Kubernetes Geneve overlay (pod network, all nodes)	Temporary	
5	Master/worker	172.17.31.128/27	Master/worker	App	172.17.31.128/27	Master/worker	30000-32767	TCP/UDP	NodePort range	Temporary	
6	Master	172.17.31.128/27	Master	App	172.17.31.128/27	Master	112 (VRRP)	VRRP	keepalived — API VIP failover (masters only)	Temporary	
7	Worker	172.17.31.128/27	Worker	App	172.17.31.128/27	Worker	112 (VRRP)	VRRP	keepalived — Ingress VIP failover (workers only)	Temporary	
8	Master/worker	172.17.31.128/27	Master/worker	App	172.17.31.128/27	Master/worker	—	ICMP	Reachability/diagnostics	Temporary	
9	Master/worker	172.17.31.128/27	Master/worker	App	172.16.26.30	h-cis-np-vcw01.ibcc.net	443	TCP	vSphere CSI (CNS) + Machine API	Temporary	
10	Master/worker	172.17.31.128/27	Master/worker	App	172.16.26.40, 172.31.23.40	NTP Server	123	UDP	Time sync	Temporary	
11	Master/worker	172.17.31.128/27	Master/worker	App	172.17.31.136	h-cis-ua-quay01	8443	TCP	Image pulls	Temporary	
12	Bastion	172.17.31.139	h-cis-ua-ocjp01	App	172.16.26.30	h-cis-np-vcw01.ibcc.net	443	TCP	Installer VM provisioning	Temporary	
13	Bastion	172.17.31.139	h-cis-ua-ocjp01	App	172.17.31.128/27	Master	6443	TCP	Post-install oc admin	Temporary	
14	Bastion	172.17.31.139	h-cis-ua-ocjp01	App	172.17.31.136	h-cis-ua-quay01	8443	TCP	Mirror registry pushes	Temporary	
15	Bastion	172.17.31.139	h-cis-ua-ocjp01	App	—	DNS Server	53	TCP/UDP	Resolving mirror, vCenter, cluster DNS records before/during install	Temporary	
16	Bastion	172.17.31.139	h-cis-ua-ocjp01	App	172.17.31.128/27	Master	22623	TCP	Bootstrap ignition checks during troubleshooting	Temporary	
17	Bastion	172.17.31.139	h-cis-ua-ocjp01	App	172.17.31.128/27	Worker	80, 443	TCP	Console/route reachability checks from the bastion (post-install validation)	Temporary	
18	Master/worker	172.17.31.128/27	Master/worker	App	172.17.31.128/27	Master/worker	1936	TCP	Router/Ingress controller stats & metrics endpoint	Temporary	
19	Master	172.17.31.128/27	Master	App	172.17.31.128/27	Bootstrap	22623	TCP	Masters fetch their own ignition config from bootstrap's temporary MCS during first boot	Temporary	Remove once bootstrap VM is destroyed
20	Bootstrap	172.17.31.128/27	Bootstrap	App	172.17.31.128/27	Master	6443	TCP	Bootstrap's temporary control plane monitors master API readiness before hand-off	Temporary	Remove once bootstrap VM is destroyed
21	Bootstrap	172.17.31.128/27	Bootstrap	App	172.17.31.128/27	Master	112 (VRRP)	VRRP	Bootstrap transiently joins the keepalived VRRP group for the API VIP	Temporary	Remove once bootstrap VM is destroyed
22	Bootstrap	172.17.31.128/27	Bootstrap	App	172.17.31.136	h-cis-ua-quay01	8443	TCP	Bootstrap pulls images to run its temporary control-plane containers (etcd, kube-apiserver, etc.)	Temporary	Remove once bootstrap VM is destroyed
23	Bootstrap	172.17.31.128/27	Bootstrap	App	—	DNS Server	53	TCP/UDP	Resolving mirror FQDN and api-int during bring-up	Temporary	Remove once bootstrap VM is destroyed
24	Bootstrap	172.17.31.128/27	Bootstrap	App	172.16.26.40, 172.31.23.40	NTP Server	123	UDP	Clock sync during bootstrap	Temporary	Remove once bootstrap VM is destroyed
25	Bastion	172.17.31.139	h-cis-ua-ocjp01	App	172.17.31.128/27	Bootstrap	22	TCP	SSH for openshift-install gather bootstrap / manual log collection	Temporary	Remove once bootstrap VM is destroyed
26	Bastion	172.17.31.139	h-cis-ua-ocjp01	App	172.17.31.128/27	Bootstrap	22623	TCP	Direct debug checks against bootstrap's MCS while troubleshooting	Temporary	Remove once bootstrap VM is destroyed
27	Master/worker	172.17.31.128/27	Master/worker	App	172.17.31.128/27	Master/worker	123	UDP	NTP peer sync between nodes (documented separately from node-to-NTP-server flow)	Temporary	
28	Admin workstation(s)	—	Admin workstation(s)	App	172.17.31.128/27	API VIP	6443	TCP	Direct oc/console API access for platform team, if not solely via bastion	Temporary	Delete row if all human access is via bastion only
29	Admin workstation(s)	—	Admin workstation(s)	App	172.17.31.128/27	Ingress VIP	80, 443	TCP	Direct web console/application access, if not solely via bastion	Temporary	Delete row if all human access is via bastion only